

KCET-2016-Set-A-1
Questions – Mathematics

1. The set A has 4 elements and the set B has 5 elements, then the number of injective mappings that can be defined from A to B is
 - (1) 144
 - (2) 72
 - (3) 60
 - (4) 120

2. Let $f : R \rightarrow R$ be defined by $f(x) = 2x + 6$, which is a bijective mapping, then $f^{-1}(x)$ is given by
 - (1) $\frac{x}{2} - 3$
 - (2) $2x + 6$
 - (3) $x - 3$
 - (4) $6x + 2$

3. Let $*$ be a binary operation defined on R by

$$a * b = \frac{a+b}{4}, \forall a, b \in R, \text{ then the operation } * \text{ is}$$

- (1) Commutative and associative
- (2) Commutative but not associative
- (3) Associative but not commutative
- (4) Neither associative nor commutative

4. The value of $\sin^{-1}\left(\cos\frac{53\pi}{5}\right)$ is

- (1) $\frac{3\pi}{5}$
- (2) $\frac{-3\pi}{5}$
- (3) $\frac{\pi}{10}$
- (4) $\frac{-\pi}{10}$

5. If $3\tan^{-1}x + \cot^{-1}x = \pi$, then x is equal to

- (1) 0

- (2) 1
- (3) -1
- (4) $1/2$

6. The simplest form of

$$\tan^{-1}\left(\frac{x}{y}\right) - \tan^{-1}\left(\frac{x-y}{x+y}\right) \text{ is equal to}$$

- (1) 0
- (2) $\frac{\pi}{4}$
- (3) $\frac{\pi}{2}$
- (4) π

7. If x, y, z are all different and not equal to zero and

$$\begin{vmatrix} 1+x & 1 & 1 \\ 1 & 1+y & 1 \\ 1 & 1 & 1+z \end{vmatrix} = 0, \text{ then the value of } x^{-1} + y^{-1} + z^{-1} \text{ is equal}$$

to

(1) xyz

(2) $x^{-1}y^{-1}z^{-1}$

(3) $-x - y - z$

(4) -1

8. If A is a square matrix of order 3×3 , then $|3A|$ is equal to

1) $3|A|$

2) $\frac{1}{3}|A|$

3) $27|A|$

4) $9|A|$

9. If $y = e^{\sin^{-1}(t^2-1)}$ and $x = e^{\sec^{-1}\left(\frac{1}{t^2-1}\right)}$ then $\frac{dy}{dx}$ is equal to

(1) $\frac{x}{y}$

(2) $\frac{-y}{x}$

$$(3) \frac{y}{x}$$

$$(4) \frac{-x}{y}$$

10. If

$$A = \frac{1}{\pi} \begin{bmatrix} \sin^{-1}(\pi x) & \tan^{-1}\left(\frac{x}{\pi}\right) \\ \sin^{-1}\left(\frac{x}{\pi}\right) & -\tan^{-1}(\pi x) \end{bmatrix}$$
$$B = \frac{1}{\pi} \begin{bmatrix} -\cos^{-1}(\pi x) & \tan^{-1}\left(\frac{x}{\pi}\right) \\ \sin^{-1}\left(\frac{x}{\pi}\right) & -\tan^{-1}(\pi x) \end{bmatrix}$$

Then $A - B$ is equal to

$$(1) I$$

$$(2) 0$$

$$(3) 2I$$

$$(4) \frac{1}{2}I$$

11. If $x^y = e^{x-y}$, then $\frac{dy}{dx}$ is equal to

(1) $\frac{\log x}{\log(x-y)}$

(2) $\frac{e^x}{x^{x-y}}$

(3) $\frac{\log x}{(1+\log x)^2}$

(4) $\frac{1}{y} - \frac{1}{x-y}$

12. If A is a matrix of order $m \times n$ and B is a matrix, such that AB' and $B'A$ are both defined, then the order of the matrix B is

(1) $m \times m$

(2) $n \times n$

(3) $n \times m$

(4) $m \times n$

13. The value of $\int \frac{e^x(1+x)}{\cos^2(e^x \cdot x)} dx$ is equal to

(1) $-\cot(ex^x) + C$

(2) $\tan(e^x \cdot x) + C$

(3) $\tan(e^x) + C$

(4) $\cot(e^x) + C$

14. If x, y, z are not equal and $\neq 0, \neq 1$, then the value of

$\begin{vmatrix} \log x & \log y & \log z \\ \log 2x & \log 2y & \log 2z \\ \log 3x & \log 3y & \log 3z \end{vmatrix}$ is equal to

(1) $\log(xyz)$

(2) $\log(6xyz)$

(3) 0

(4) $\log(x + y + z)$

15. The function $f(x) = [x]$ where $[x]$ is the greatest integer function, is continuous at

(1) 1.5

- (2) 4
- (3) 1
- (4) -2

16. The value of $\int \frac{e^x (x^2 \tan^{-1} x + \tan^{-1} x + 1)}{x^2 + 1} dx$ is equal to

- (1) $e^x \tan^{-1} x + C$
- (2) $\tan^{-1}(e^x) + C$
- (3) $\tan^{-1}(x^e) + C$
- (4) $e^{\tan^{-1} x} + C$

17. If $2\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| \cdot |\mathbf{b}|$, then the angle between $\mathbf{a}$ and $\mathbf{b}$ is

- (1) 30°
- (2) 0°
- (3) 90°
- (4) 60°

18. If $x^m y^n = (x + y)^{m+n}$, then $\frac{dy}{dx}$ is equal to

(1) $\frac{x+y}{xy}$

(2) xy

(3) 0

(4) $\frac{y}{x}$

19. The general solution of $\cot \theta + \tan \theta = 2$ is

(1) $\theta = \frac{n\pi}{2} + (-1)^n \frac{\pi}{8}$

(2) $\theta = \frac{n\pi}{2} + (-1)^n \frac{\pi}{4}$

(3) $\theta = \frac{n\pi}{2} + (-1)^n \frac{\pi}{6}$

(4) $\theta = n\pi + (-1)^n \frac{\pi}{8}$

20. The value of $\int_{-\pi/4}^{\pi/4} \sin^{103} x \cdot \cos^{101} x \, dx$

(1) $\left(\frac{\pi}{2}\right)^{103}$

(2) $\left(\frac{\pi}{4}\right)^{101}$

(3) 2

(4) 0

21. The length of the latus rectum of the parabola $4y^2 + 3x + 3y + 1 = 0$

(1) $4/3$

(2) 7

(3) 12

(4) $3/4$

22. The value of $\int \frac{e^{6\log x} - e^{5\log x}}{e^{4\log x} - e^{3\log x}} dx$ is equal to

(1) $0 + C$

(2) $\frac{x^3}{3} + C$

(3) $\frac{3}{x^3} + C$

(4) $\frac{1}{x} + C$

23. The differential coefficient of $\log_{10} x$ with respect to $\log_x 10$ is

(1) 1

(2) $-(\log_{10} x)^2$

(3) $(\log_x 10)^2$

(4) $\frac{x^2}{100}$

24. The slope of the tangent to the curve $x = t^2 + 3t - 8$,
 $y = 2t^2 - 2t - 5$ at the point $(2, -1)$ is

(1) $\frac{22}{7}$

(2) $\frac{6}{7}$

(3) $\frac{7}{6}$

(4) $\frac{-6}{7}$

25. The real part of $(1 - \cos\theta + i\sin\theta)^{-1}$ is

(1) $\frac{1}{2}$

(2) $\frac{1}{1 + \cos\theta}$

(3) $\tan\frac{\theta}{2}$

(4) $\cot\frac{\theta}{2}$

26. $\int_0^{\pi/2} \frac{\sin^{1000} x}{\sin^{1000} x + \cos^{1000} x} dx$ is equal to

(1) 1000

(2) 1

(3) $\frac{\pi}{2}$

(4) $\frac{\pi}{4}$

27. If $1 + \sin \theta + \sin^2 \theta + \dots$ upto $\infty = 2\sqrt{3} + 4$, then θ is equal to

(1) $\frac{\pi}{6}$

(2) $\frac{\pi}{4}$

(3) $\frac{\pi}{3}$

(4) $\frac{3\pi}{4}$

28. $\lim_{x \rightarrow 0} \frac{xe^x - \sin x}{x}$ is equal to

(1) 3

(2) 1

(3) 0

(4) 2

29. If $\tan^{-1}(x^2 + y^2) = \alpha$, then $\frac{dy}{dx}$ is equal to

(1) $\frac{-x}{y}$

(2) xy

(3) $\frac{x}{y}$

(4) $-xy$

30. The simplified form of $i^n + i^{n+1} + i^{n+2} + i^{n+3}$ is

(1) 0

(2) 1

(3) -1

(4) i

31. The two curves $x^3 - 3xy^2 + 2 = 0$ and $3x^2y - y^3 = 2$

(1) touch each other

(2) cut each other at right angle

(3) cut at an angle $\pi/3$

(4) cut at an angle $\pi/4$

32. The equation of the normal to the curve $y(1+x^2)=2-x$, where the tangent crosses X -axis, is

(1) $5x - y - 10 = 0$

(2) $x - 5y - 10 = 0$

(3) $5x + y - 10 = 0$

(4) $x + 5y + 10 = 0$

33. The maximum value of $\left(\frac{1}{x}\right)^x$ is

(1) e

(2) e^e

(3) $e^{1/e}$

(4) $\left(\frac{1}{e}\right)^e$

34. The solution of the differential equation $\frac{dy}{y} + \frac{dx}{x} = 0$ is

$$(1) \frac{1}{y} + \frac{1}{x} = C$$

$$(2) \log x \cdot \log y = C$$

$$(3) xy = C$$

$$(4) x + y = C$$

35. The order and degree of the differential equation

$$\left[1 + \left(\frac{dy}{dx} \right)^2 + \sin \left(\frac{dy}{dx} \right) \right]^{3/4} = \frac{d^2 y}{dx^2}$$

$$(1) \text{ Order} = 2, \text{ degree} = 3$$

$$(2) \text{ Order} = 2, \text{ degree} = 4$$

$$(3) \text{ Order} = 2, \text{ degree} = \frac{3}{4}$$

$$(4) \text{ Order} = 2, \text{ degree} = \text{not defined}$$

36. If **a** and **b** are the unit vectors, then what is the angle between **a** and **b** for $\sqrt{3}\mathbf{a} - \mathbf{b}$ to be unit vector?

$$(1) 30^\circ$$

$$(2) 45^\circ$$

(3) 60°

(4) 90°

37. The sum of 1st n terms of the series

$$\frac{1^2}{1} + \frac{1^2 + 2^2}{1 + 2} + \frac{1^2 + 2^2 + 3^2}{1 + 2 + 3} + \dots \text{ is}$$

(1) $\frac{n+2}{3}$

(2) $\frac{n(n+2)}{3}$

(3) $\frac{n(n-2)}{3}$

(4) $\frac{n(n-2)}{6}$

38. The 11th term in the expansion of $\left(x + \frac{1}{\sqrt{x}}\right)^{14}$

(1) $\frac{999}{x}$

(2) $\frac{1001}{x}$

(3) i

(4) $\frac{x}{1001}$

39. Suppose $\mathbf{a} + \mathbf{b} + \mathbf{c} = 0$, $|\mathbf{a}| = 3$, $|\mathbf{b}| = 5$, $|\mathbf{c}| = 7$ then the angle between $\mathbf{a}$ and $\mathbf{b}$ is

(1) π

(2) $\frac{\pi}{2}$

(3) $\frac{\pi}{3}$

(4) $\frac{\pi}{4}$

40. If $\mathbf{a} = 3$, $\mathbf{b} = 4$, $\mathbf{c} = 5$ each one is perpendicular to the sum of the remaining then $|\mathbf{a} + \mathbf{b} + \mathbf{c}|$ is equal to

(1) $\frac{5}{\sqrt{2}}$

(2) $\frac{2}{\sqrt{5}}$

(3) $5\sqrt{2}$

(4) $\sqrt{5}$

41. If a straight lines $2x + 3y - 3 = 0$ and $x + ky + 7 = 0$ are perpendicular, then the value of k is

(1) $\frac{2}{3}$

(2) $\frac{3}{2}$

(3) $\frac{-2}{3}$

(4) $\frac{-3}{2}$

42. The rate of change of area of a circle with respect to its radius at $r = 2$ cm is

(1) 4

(2) 2π

- (3) 2
- (4) 4π

43. The value of $\tan \frac{\pi}{8}$ is equal to

- (1) $\frac{1}{2}$
- (2) $\sqrt{2} + 1$
- (3) $\frac{1}{\sqrt{2} + 1}$
- (4) $1 - \sqrt{2}$

44. Area lying between the curves $y^2 = 2x$ and $y = x$ is

- (1) $\frac{2}{3}$ sq unit
- (2) $\frac{1}{3}$ sq unit
- (3) $\frac{1}{4}$ sq unit
- (4) $\frac{3}{4}$ sq unit

45. If $P(A \cap B) = \frac{7}{10}$ and $P(B) = \frac{17}{20}$, where P stands for probability, then $P(A/B)$ is equal to

(1) $\frac{7}{8}$

(2) $\frac{17}{20}$

(3) $\frac{14}{17}$

(4) $\frac{1}{8}$

46. The coefficient of variation of two distributions are 60 and 70. The standard deviations are 21 and 16 respectively, then their mean is

(1) 35

(2) 23

(3) 28.25

(4) 22.85

47. Two cards are drawn at random from a pack of 52 cards. The probability of these two being 'Aces' is

(1) $\frac{1}{26}$

(2) $\frac{1}{221}$

(3) $\frac{1}{2}$

(4) $\frac{1}{13}$

48. If $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$, then x^2 is equal to

(1) $1 - y^2$

(2) y^2

(3) 0

(4) $\sqrt{1 - y}$

49. The value of $\int_2^8 \frac{\sqrt{10-x}}{\sqrt{x} + \sqrt{10-x}} dx$ is

(1) 10

(2) 0

(3) 8

(4) 3

50. The contrapositive of the converse of the statement “If x is a prime number, then x is odd”, is

(1) If x is not prime number, then x is odd

(2) If x is not an odd number, then x is not a prime number

(3) If x is a prime number, then it is not odd.

(4) If x is not a prime number, then x is not an odd.

51. Two dice are thrown simultaneously, the probability of obtaining a total score of 5 is

(1) $\frac{1}{18}$

(2) $\frac{1}{12}$

(3) $\frac{1}{9}$

(4) $\frac{1}{6}$

52. If $A = \begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}$ and $A + A^T = I$, where I is the unit matrix of 2×2 and A^T is the transpose of A , then the value of θ is equal to

(1) $\frac{\pi}{6}$

(2) $\frac{\pi}{3}$

(3) π

(4) $\frac{3\pi}{2}$

53. If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, then $A^2 - 5A$ is equal to

(1) I

(2) $-I$

(3) $7I$

(4) $-7I$

54. The value of x , if $x(\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}})$ is a unit vector, is

(1) $\pm \frac{1}{\sqrt{3}}$

(2) $\pm \sqrt{3}$

(3) ± 3

(4) $\pm \frac{1}{3}$

55. If $x = 2 + 3\cos\theta$ and $y = 1 - 3\sin\theta$ represent a circle, then the centre and radius is

(1) $(2,1) 9$

(2) $(2,1) 3$

(3) $(1,2) \frac{1}{3}$

(4) $(-2,-1) 3$

56. The vector equation of the plane, which is at a distance of $\frac{3}{\sqrt{14}}$

from the origin and the normal from the origin is $2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + \hat{\mathbf{k}}$ is

(1) $\mathbf{r} \cdot (2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + \hat{\mathbf{k}}) = 3$

(2) $\mathbf{r} \cdot (\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}) = 9$

(3) $\mathbf{r} \cdot (\hat{\mathbf{i}} + 2\hat{\mathbf{j}}) = 3$

(4) $\mathbf{r} \cdot (2\hat{\mathbf{i}} + \hat{\mathbf{k}}) = 3$

57. Find the coordinates of the foot of the perpendicular drawn from the origin to the plane $5y + 8 = 0$.

(1) $\left(0, -\frac{18}{5}, 2\right)$

(2) $\left(0, \frac{8}{5}, 0\right)$

(3) $\left(\frac{8}{25}, 0, 0\right)$

(4) $\left(0, -\frac{8}{5}, 0\right)$

58. If $\cos \alpha, \cos \beta, \cos \gamma$ are the direction cosines of a vector $\mathbf{a}$, then $\cos 2\alpha + \cos 2\beta + \cos 2\gamma$ is equal to

- (1) 2
- (2) 3
- (3) -1
- (4) 0

59. The value of the $\sin 1^\circ + \sin 2^\circ + \dots + \sin 359^\circ$ is equal to

- (1) 0
- (2) 1
- (3) -1
- (4) 180

60. Integrating factor of $x \frac{dy}{dx} - y = x^4 - 3x$ is

- (1) x
- (2) $\log x$
- (3) $\frac{1}{x}$
- (4) $-x$

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Answer keys

1	4	16	1	31	1	46	4
2	1	17	4	32	1	47	2
3	2	18	4	33	3	48	1
4	4	19	2	34	3	49	4
5	2	20	4	35	4	50	4
6	2	21	4	36	1	51	3
7	4	22	2	37	2	52	1
8	3	23	2	38	2	53	4
9	2	24	2	39	3	54	1
10	4	25	1	40	3	55	2
11	3	26	4	41	4	56	1
12	4	27	3	42	4	57	4
13	2	28	3	43	3	58	3
14	1	29	1	44	1	59	1
15	1	30	1	45	3	60	3

KCET-2016-Set-A-1
Solutions – Mathematics

1. The total number of injective mapping from set A to set B is calculated as,

$$\begin{aligned}\text{Number of mapping} &= {}^5P_4 \\ &= \frac{5!}{1!} \\ &= 120\end{aligned}$$

2. The function given is $f(x) = 2x + 6$

The value of $f^{-1}(x)$ calculates as,

We know that,

$$f\{f^{-1}(x)\} = x$$

$$2f^{-1}(x) + 6 = x$$

$$\begin{aligned}f^{-1}(x) &= \frac{x-6}{2} \\ &= \frac{x}{2} - 3\end{aligned}$$

3. According to the question,

$$a * b = \frac{a + b}{4}$$

According to the commutative property,

$$\frac{a + b}{4} = \frac{b + a}{4}$$

Which implies the given operation is commutative in nature.

According to the associative property,

$$a * (b * c) = (a * b) * c$$

$$\frac{a + \frac{b + c}{4}}{4} = \frac{\left(\frac{a + b}{4}\right) + c}{4}$$

$$\text{L.H.S} \neq \text{R.H.S}$$

Which implies that $*$ binary operation is not associative.

4. The value of $\sin^{-1}\left(\cos\frac{53\pi}{5}\right)$ is calculated as,

$$3 \tan^{-1} x + \cot^{-1} x = \pi$$

$$2 \tan^{-1} x + \tan^{-1} x + \cot^{-1} x = \pi$$

$$2 \tan^{-1} x + \frac{\pi}{2} = \pi$$

$$2 \tan^{-1} x = \pi - \frac{\pi}{2}$$

$$2 \tan^{-1} x = \frac{\pi}{2}$$

$$\tan^{-1} x = \frac{\pi}{4}$$

$$x = \tan\left(\frac{\pi}{4}\right)$$

$$\begin{aligned}\sin^{-1}\left(\cos\frac{53\pi}{5}\right) &= \sin^{-1}\left[\cos\left(10\pi + \frac{3\pi}{5}\right)\right] \\ &= \sin^{-1}\left[\cos\left(\frac{3\pi}{5}\right)\right] \\ &= \sin^{-1}\left[\sin\left(\frac{\pi}{2} - \frac{3\pi}{5}\right)\right] \\ &= \sin^{-1}\left[\sin\left(\frac{5\pi - 6\pi}{10}\right)\right] \\ &= \sin^{-1}\left[\sin\left(\frac{-\pi}{10}\right)\right] \\ &= \frac{-\pi}{10}\end{aligned}$$

5. The given expression can be simplified as,

$$\begin{aligned}
 \tan^{-1}\left(\frac{x}{y}\right) - \tan^{-1}\left(\frac{x-y}{x+y}\right) &= \tan^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{x-y}{x+y}\right) \\
 &= \tan^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{1-x/y}{1+x/y}\right) \\
 &= \tan^{-1}\left(\frac{x}{y}\right) + \tan^{-1}(1) - \tan^{-1}\left(\frac{x}{y}\right) \\
 &= \tan^{-1}(1) \\
 &= \frac{\pi}{4}
 \end{aligned}$$

6. The given matrix is $\begin{vmatrix} 1+x & 1 & 1 \\ 1 & 1+y & 1 \\ 1 & 1 & 1+z \end{vmatrix} = 0$

Expand the given matrix along the R_1 , we get

$$(1+x)[(1+y)(1+z)-1] - 1(1+z-1) + (1-1-y) = 0$$

$$(1+x)(1+y)(1+z) - (1+x) - z - y = 0$$

$$(1+x)(1+y)(1+z) = x + y + z + 1$$

$$1 + x + y + z + xy + yz + xz + xyz = x + y + z + 1$$

$$xy + yz + xz = -xyz$$

Divide both side by xyz .

$$\frac{1}{z} + \frac{1}{x} + \frac{1}{y} = -1$$

$$x^{-1} + y^{-1} + z^{-1} = -1$$

7. According to the property of square matrix of order n

$$|KA| = K^n |A|$$

$$\begin{aligned} |3A| &= 3^3 |A| \\ &= 27 |A| \end{aligned}$$

9. Now, $y = e^{\sin^{-1}(t^2-1)}$ and $x = e^{\sec^{-1}\left(\frac{1}{t^2-1}\right)}$.

The differential equation is solved as,

$$xy = e^{\sin^{-1}(t^2-1)} \cdot e^{\sec^{-1}\left(\frac{1}{t^2-1}\right)}$$

Use the identity $\sec^{-1}\left(\frac{1}{x}\right) = \cos^{-1}(x)$.

$$xy = e^{\sin^{-1}(t^2-1)} \cdot e^{\sec^{-1}\left(\frac{1}{t^2-1}\right)}$$

$$xy = e^{\sin^{-1}(t^2-1) + \cos^{-1}(t^2-1)}$$

$$xy = e^{\frac{\pi}{2}}$$

Differentiate the equation on both sides.

$$x \frac{dy}{dx} + y = 0$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

10. The following operation of the matrix is calculated as,

$$\begin{aligned} A - B &= \frac{1}{\pi} \begin{bmatrix} \sin^{-1}(\pi x) + \cos^{-1}(\pi x) & \tan^{-1}\left(\frac{x}{\pi}\right) - \tan^{-1}\left(\frac{x}{\pi}\right) \\ \sin^{-1}\left(\frac{x}{\pi}\right) - \sin^{-1}\left(\frac{x}{\pi}\right) & \cos^{-1}(\pi x) + \sin^{-1}(\pi x) \end{bmatrix} \\ &= \frac{1}{\pi} \begin{bmatrix} \frac{\pi}{2} & 0 \\ 0 & \frac{\pi}{2} \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \\ &= \frac{1}{2} I \end{aligned}$$

11. The equation is $x^y = e^{x-y}$.

Take log on both the sides of the given equation.

$$y \log x = x - y$$

$$y \log x + y = x$$

$$y = \frac{x}{1 + \log x}$$

Differentiate the equation on both the sides.

$$\begin{aligned} \frac{dy}{dx} &= \frac{(1 + \log x)(1) - x \left(\frac{1}{x} \right)}{(1 + \log x)^2} \\ &= \frac{1 + \log x - 1}{(1 + \log x)^2} \\ &= \frac{\log x}{(1 + \log x)^2} \end{aligned}$$

12. Sol: The given order of the matrix is $m \times n$.

Let the order of the matrix B be $u \times v$.

Which implies that the order of the matrix B' is $v \times u$.

Now, the matrix AB' is defined as

Column of A = Row of B'

$$n = v$$

Now, the matrix $B'A$ is defined as

Column of $B' =$ Row of A

$$u = m$$

Which implies that the order of the matrix B is $m \times n$.

13. The given integration is as,

$$I = \int \frac{e^x(1+x)}{\cos^2(e^x \cdot x)} dx$$

Substitute $e^x \cdot x = t$ in the given integration.

$$e^x \cdot x = t$$

$$e^x(x+1)dx = dt$$

$$e^x(x+1)dx = dt$$

The integration is solved as,

$$I = \int \frac{e^x(1+x)}{\cos^2(e^x \cdot x)} dx$$

$$= \int \frac{dt}{\cos^2 t}$$

$$= \int \sec^2 t dt$$

$$= \tan t + C$$

$$= \tan(e^x \cdot x) + C$$

14. The given matrix is
$$\begin{vmatrix} \log x & \log y & \log z \\ \log 2x & \log 2y & \log 2z \\ \log 3x & \log 3y & \log 3z \end{vmatrix}.$$

Apply the following operation $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$.

$$= \begin{vmatrix} \log x & \log y & \log z \\ \log 2x - \log x & \log 2y - \log y & \log 2z - \log z \\ \log 3x - \log x & \log 3y - \log y & \log 3z - \log z \end{vmatrix}$$

$$= \begin{vmatrix} \log x & \log y & \log z \\ \log 2 & \log 2 & \log 2 \\ \log 3 & \log 3 & \log 3 \end{vmatrix}$$

$$= \log 2 \cdot \log 3 \begin{vmatrix} \log x & \log y & \log z \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

As R_2 and R_3 are same.

The value of the determinant is zero.

15. The given function is $f(x) = [x]$ which is continuous at only non integral value.

Which implies that $f(x)$ is continuous at $x = 1.5$.

16. Let assume that

$$\begin{aligned} I &= \int e^x \left(\frac{x^2 \tan^{-1} x + \tan^{-1} x + 1}{x^2 + 1} \right) dx \\ &= \int e^x \left(\tan^{-1} x + \frac{1}{x^2 + 1} \right) dx \end{aligned}$$

If $f(x) = \tan^{-1} x$, differentiate it as,

$$f'(x) = \frac{1}{x^2 + 1}$$

The given integration is solved as,

$$\begin{aligned} I &= \int e^x \left(\tan^{-1} x + \frac{1}{x^2 + 1} \right) dx \\ &= e^x \tan^{-1} x + C \end{aligned}$$

17. The given equation is $2\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| \cdot |\mathbf{b}|$.

Condition for the angle between the non zero vector is as follows,

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| \cdot |\mathbf{b}| \cos \theta$$

$$2|\mathbf{a}| \cdot |\mathbf{b}| \cos \theta = |\mathbf{a}| \cdot |\mathbf{b}|$$

$$\cos \theta = \frac{1}{2}$$

$$\theta = 60^\circ$$

18. The given equation is $x^m y^n = (x + y)^{m+n}$.

Taking log on both sides,

$$m \log x + n \log y = (m + n) \log(x + y)$$

Differentiate the equation on both sides.

$$\frac{m}{n} + \frac{n}{y} \cdot \frac{dy}{dx} = \frac{m + n}{x + y} \left(1 + \frac{dy}{dx} \right)$$

$$\frac{n}{y} \cdot \frac{dy}{dx} - \left(\frac{m + n}{x + y} \right) \frac{dy}{dx} = \frac{m + n}{x + y} - \frac{m}{x}$$

$$\frac{dy}{dx} \left(\frac{n}{y} - \frac{(m + n)}{x + y} \right) = \frac{m + n}{x + y} - \frac{m}{x}$$

$$\frac{dy}{dx} \left(\frac{nx + ny - my - ny}{y(x + y)} \right) = \frac{mx + nx - mx - my}{x(x + y)}$$

Further solve,

$$\frac{dy}{dx} \left(\frac{nx - my}{y(x + y)} \right) = \frac{nx - my}{x(x + y)}$$

$$\frac{dy}{dx} = \frac{y}{x}$$

19. The solution of the given equation is calculated as,

$$\cot \theta + \tan \theta = 2$$

$$\frac{\cos \theta}{\sin \theta} + \frac{\sin \theta}{\cos \theta} = 2$$

$$\frac{\cos^2 \theta + \sin^2 \theta}{\sin \theta \cos \theta} = 2$$

$$1 = 2 \sin \theta \cos \theta$$

Solve further for the solution as,

$$1 = 2 \sin \theta \cos \theta$$

$$\sin 2\theta = 1$$

$$\sin 2\theta = \sin \frac{\pi}{2}$$

$$\theta = \frac{n\pi}{2} + (-1)^n \frac{\pi}{4}$$

20. Let assume that $I = \int_{-\pi/4}^{\pi/4} \sin^{103} x \cdot \cos^{101} x dx$.

If $f(x) = \sin^{103} x \cdot \cos^{101} x$ then,

$$f(-x) = \sin^{103}(-x) \cos^{101}(-x)$$

$$= -\sin^{103} x \cos^{101} x$$

$$= -f(x)$$

This implies that $I = 0$.

21. The given equation of parabola is $4y^2 + 3x^2 + 3y + 1 = 0$.

The equation can be expressed as,

$$4y^2 + 3x + 3y + 1 = 0$$

$$4y^2 + 3y = -3x - 1$$

$$y^2 + \frac{3}{4}y = -\frac{3}{4}x - \frac{1}{4}$$

$$y^2 + \frac{3}{4}y + \frac{9}{64} = -\frac{3}{4}x - \frac{1}{4} + \frac{9}{64}$$

Further solve it for solution as,

$$y^2 + \frac{3}{4}y + \frac{9}{64} = -\frac{3}{4}x - \frac{1}{4} + \frac{9}{64}$$

$$\left(y + \frac{3}{8}\right)^2 = -\frac{3}{4}\left(x + \frac{7}{48}\right)$$

Which implies that the length of latus rectum is $\frac{3}{4}$.

22. The given integration is solved as,

$$\begin{aligned}
 I &= \int \frac{e^{6\log x} - e^{5\log x}}{e^{4\log x} - e^{3\log x}} dx \\
 &= \int \frac{x^6 - x^5}{x^4 - x^3} dx \\
 &= \int \frac{x^5(x-1)}{x^3(x-1)} dx \\
 &= \int x^2 dx \\
 &= \frac{x^3}{3} + C
 \end{aligned}$$

23. Let assume that $u = \log_{10} x$ and $v = \log_x 10$.

$$u = \frac{\log_e x}{\log_e 10} \text{ and } v = \frac{\log_e 10}{\log_e x}$$

$$\frac{du}{dx} = \frac{1}{x \log_e 10} \text{ and } \frac{dv}{dx} = \log_e 10 \left[\frac{-1}{x(\log_e x)^2} \right]$$

$$\frac{du}{dv} = \frac{du/dx}{dv/dx}$$

$$= \frac{1}{x \log_e 10} \div \frac{-\log_e 10}{x(\log_e x)^2}$$

Solve further,

$$\begin{aligned}\frac{du}{dv} &= \frac{-(\log_e x)^2}{(\log_e 10)^2} \\ &= -(\log_{10} x)^2\end{aligned}$$

24. Given equations $x = t^2 + 3t - 8$ and $y = 2t^2 - 2t - 5$.

Substitute $x = 2$.

$$2 = t^2 + 3t - 8$$

$$t^2 + 3t - 10 = 0$$

$$(t + 5)(t - 2) = 0$$

$$t = -5, 2$$

Substitute $y = -1$.

$$-1 = 2t^2 - 2t - 5$$

$$2t^2 - 2t - 4 = 0$$

$$t^2 - t - 2 = 0$$

$$(t - 2)(t + 1) = 0$$

$$t = 2, -1$$

As $t \neq -1$,

This implies $t = 2$.

Solve further,

$$\frac{dx}{dt} = 2t + 3 \text{ and } \frac{dy}{dt} = 4t - 2$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

$$= \frac{4t - 2}{2t + 3}$$

$$\left(\frac{dy}{dx} \right)_{t=2} = \frac{4(2) - 2}{2(2) + 3}$$

$$= \frac{6}{7}$$

Which implies that the slope of the tangent is $\frac{6}{7}$.

25. To obtain the real sole it as,

$$\begin{aligned} (1 - \cos \theta + i \sin \theta)^{-1} &= \frac{1}{1 - \cos \theta + i \sin \theta} \\ &= \frac{1 - \cos \theta - i \sin \theta}{(1 - \cos \theta)^2 + \sin^2 \theta} \\ &= \frac{1 - \cos \theta - i \sin \theta}{1 - 2 \cos \theta + \cos^2 \theta + \sin^2 \theta} \\ &= \frac{1 - \cos \theta - i \sin \theta}{2 - 2 \cos \theta} \end{aligned}$$

Solve further,

$$\begin{aligned}
 (1 - \cos \theta + i \sin \theta)^{-1} &= \frac{1 - \cos \theta}{2(1 - \cos \theta)} - \frac{i \sin \theta}{2(1 - \cos \theta)} \\
 &= \frac{1}{2} - \frac{i \sin \theta}{2(1 - \cos \theta)}
 \end{aligned}$$

Which implies the real part is $\frac{1}{2}$.

26. The given integration is solved as,

$$I = \int_0^{\pi/2} \frac{\sin^{1000} x}{\sin^{1000} x + \cos^{1000} x} dx \dots (1)$$

$$I = \int_0^{\pi/2} \frac{\sin^{1000} \left(\frac{\pi}{2} - x \right)}{\sin^{1000} \left(\frac{\pi}{2} - x \right) + \cos^{1000} \left(\frac{\pi}{2} - x \right)} dx$$

$$I = \int_0^{\pi/2} \frac{\cos^{1000} x}{\cos^{1000} x + \sin^{1000} x} dx \dots (2)$$

Add equation (1) and (2).

$$2I = \int_0^{\pi/2} \frac{\sin^{1000} x + \cos^{1000} x}{\sin^{1000} x + \cos^{1000} x} dx$$

$$2I = \int_0^{\pi/2} 1 dx$$

$$2I = [x]_0^{\pi/2}$$

$$I = \frac{\pi}{4}$$

27. The given series is $1 + \sin \theta + \sin^2 \theta + \dots \infty = 2\sqrt{3} + 4$.

Use the following identity $1 + r + r^2 + \dots \infty = \frac{1}{1-r}$

$$\frac{1}{1 - \sin \theta} = 2\sqrt{3} + 4$$

$$\sin \theta = 1 - \frac{1}{2\sqrt{3} + 4}$$

$$\sin \theta = \frac{2\sqrt{3} + 4 - 1}{2\sqrt{3} + 4}$$

$$\sin \theta = \frac{\sqrt{3}}{2}$$

Solve further to obtain the value of the angle,

$$\sin \theta = \frac{\sqrt{3}}{2}$$

$$\theta = \frac{\sqrt{3}}{2}$$

$$= \frac{\pi}{3}$$

28. The following limit is solved as,

Use the L' Hospital rule as,

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{xe^x - \sin x}{x} &= \lim_{x \rightarrow 0} \frac{xe^x + e^x - \cos x}{1} \\ &= 0 + 1 - 1 \\ &= 0\end{aligned}$$

29. The given differential equation is $\tan^{-1}(x^2 + y^2) = \alpha$.

This implies $x^2 + y^2 = \tan \alpha$.

Differentiate the equation on both the sides.

$$2x + 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-x}{y}$$

30. The given equation is simplified as,

$$\begin{aligned}i^n + i^{n+1} + i^{n+2} + i^{n+3} &= i^n (1 + i + i^2 + i^3) \\&= i^n (1 + i - 1 - i) \\&= i^n \times 0 \\&= 0\end{aligned}$$

31. The given equation is $x^3 - 3xy^2 + 2 = 0$.

The equation is solved as,

$$x^3 - 3xy^2 + 2 = 0$$

$$3x^2 - 6xy \frac{dy}{dx} - 3y^2 = 0$$

$$\frac{dy}{dx} = \frac{3(x^2 - y^2)}{6xy}$$

$$\begin{aligned}\text{Now, } \left(\frac{dy}{dx} \right)_{(h,k)} &= \frac{3(h^2 - k^2)}{6hk} \\&= m_1\end{aligned}$$

and $3x^2y - y^3 = 2$.

$$3x^2 \frac{dy}{dx} + 6xy - 3y^2 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-6xy}{3(x^2 - y^2)}$$

$$\begin{aligned}\text{Now, } \left(\frac{dy}{dx} \right)_{(h,k)} &= \frac{-6hk}{3(h^2 - k^2)} \\ &= m_2\end{aligned}$$

$$\begin{aligned}m_1 \cdot m_2 &= \frac{3(h^2 - k^2)}{6hk} \times \frac{-6hk}{3(h^2 - k^2)} \\ &= -1\end{aligned}$$

Therefore, both the curves cut each other at right angle.

32. The given equation $y(1 + x^2) = 2 - x$.

Substitute $y = 0$ which gives $x = 2$.

Differentiate the equation on both sides.

$$\begin{aligned}\frac{dy}{dx}(1 + x^2) + 2xy &= -1 \\ \frac{dy}{dx} &= \frac{-1 - 2xy}{1 + x^2} \\ \left(\frac{dy}{dx} \right)_{(2,0)} &= \frac{-1 - 0}{1 + 4} \\ &= \frac{-1}{5}\end{aligned}$$

This implies that the slope of the normal is 5.

The equation of the normal at $(2, 0)$ is calculated as,

$$y - 0 = 5(x - 2)$$

$$y = 5x - 10$$

$$5x - y - 10 = 0$$

33. Let us assume $y = \left(\frac{1}{x}\right)^x$.

This implies $y = x^{-x}$.

Differentiate it on both the sides.

$$\frac{dy}{dx} = x^{-x}(-1 - \log x)$$

$$\frac{dy}{dx} = -x^{-x}(1 + \log x)$$

Use the following properties of function to solve it further,

$$\frac{d}{dx} f(x)^{g(x)} = f(x)^{g(x)} \text{ and}$$

$$g(x) \cdot \frac{1}{f(x)} \cdot f'(x) + g'(x) \log f(x)$$

The maxima and minima of the function is calculated as,

$$\frac{dy}{dx} = 0$$

$$1 + \log x = 0$$

$$\log x = -1$$

$$x = e^{-1}$$

Which implies the maximum value is $e^{1/e}$ at $x = e^{-1}$.

34. The given relation is $\frac{dy}{y} + \frac{dx}{x} = 0$.

Integrate both the sides.

$$\log y + \log x = \log C$$

$$\log xy = \log C$$

$$xy = C$$

35. The given differential equation is

$$\left[1 + \left(\frac{dy}{dx} \right)^2 + \sin \left(\frac{dy}{dx} \right) \right]^{3/4} = \frac{d^2 y}{dx^2}$$

It is further solved as,

$$\left(\frac{d^2 y}{dx^2} \right) = \left[1 + \left(\frac{dy}{dx} \right)^2 + \sin \left(\frac{dy}{dx} \right) \right]^3$$

Which implies order = 2 and degree not defined.

36. The given relationship is $|\mathbf{a}| = |\mathbf{b}| = |\sqrt{3}\mathbf{a} - \mathbf{b}| = 1$

$$|\sqrt{3}\mathbf{a} - \mathbf{b}|^2 = (\sqrt{3}\mathbf{a} - \mathbf{b}) \cdot (\sqrt{3}\mathbf{a} - \mathbf{b})$$

$$1 = 3|\mathbf{a}|^2 + |\mathbf{b}|^2 - 2\sqrt{3}|\mathbf{a}||\mathbf{b}|\cos\theta$$

$$1 = 3 + 1 - 2\sqrt{3}\cos\theta$$

$$2\sqrt{3}\cos\theta = 3$$

Further solve it to obtain the angle as,

$$2\sqrt{3}\cos\theta = 3$$

$$\cos\theta = \frac{\sqrt{3}}{2}$$

$$\theta = 30^\circ$$

37. The given expression is $\frac{1^2}{1} + \frac{1^2 + 2^2}{1 + 2} + \frac{1^2 + 2^2 + 3^2}{1 + 2 + 3} + \dots$

Further solve it as,

$$a_n = \frac{1^2 + 2^2 + 3^2 + \dots + n^2}{1 + 2 + 3 + \dots + n}$$

$$a_n = \frac{\frac{n(n+1)(2n+1)}{6}}{\frac{n(n+1)}{2}}$$

$$a_n = \frac{2n+1}{3}$$

The value of S_n is calculated as,

$$\begin{aligned} S_n &= \sum \frac{2n+1}{3} \\ &= \frac{1}{3} \left[\sum 2n + \sum 1 \right] \\ &= \frac{1}{3} \left[\frac{2n(n+1)}{2} + n \right] \end{aligned}$$

Further solve it as,

$$\begin{aligned} S_n &= \frac{1}{3} [n^2 + n + n] \\ &= \frac{1}{3} [n^2 + 2n] \\ &= \frac{n(n+2)}{3} \end{aligned}$$

38. The given expression is $\left(x + \frac{1}{\sqrt{x}}\right)^{14}$.

The value of T_{11} is calculated as,

$$\begin{aligned} T_{11} &= {}^{14}C_{10} (x)^4 \left(\frac{1}{\sqrt{x}}\right)^{10} \\ &= \frac{14!}{4!10!} \cdot \frac{x^4}{x^5} \\ &= \frac{14 \times 13 \times 12 \times 11}{1 \times 2 \times 3 \times 4} \cdot \frac{1}{x} \\ &= \frac{1001}{x} \end{aligned}$$

39. The given equation is $\mathbf{a} + \mathbf{b} + \mathbf{c} = 0$ and $|\mathbf{a}| = 3, |\mathbf{b}| = 5, |\mathbf{c}| = 7$

$$\mathbf{a} + \mathbf{b} + \mathbf{c} = 0$$

$$\mathbf{a} + \mathbf{b} = -\mathbf{c}$$

$$|\mathbf{a} + \mathbf{b}| = |-\mathbf{c}|^2$$

$$(\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} + \mathbf{b}) = 49$$

Solve it further as,

$$|\mathbf{a}|^2 + |\mathbf{b}|^2 + 2|\mathbf{a}||\mathbf{b}|\cos\theta = 49$$

$$9 + 25 + 2 \times 3 \times 5 \times \cos\theta = 49$$

$$\cos\theta = \frac{49 - 25 - 9}{30}$$

$$\cos\theta = \frac{1}{2}$$

$$\theta = 60^\circ$$

$$= \frac{\pi}{3}$$

40. The given values are

$$|\mathbf{a}| = 3, |\mathbf{b}| = 4, |\mathbf{c}| = 5 \text{ and } \mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a} = 0$$

Now,

$$|\mathbf{a} + \mathbf{b} + \mathbf{c}|^2 = (\mathbf{a} + \mathbf{b} + \mathbf{c}) \cdot (\mathbf{a} + \mathbf{b} + \mathbf{c})$$

$$|\mathbf{a} + \mathbf{b} + \mathbf{c}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 + |\mathbf{c}|^2 + 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a})$$

$$|\mathbf{a} + \mathbf{b} + \mathbf{c}|^2 = (3)^2 + (4)^2 + (5)^2 + 2(0)$$

$$|\mathbf{a} + \mathbf{b} + \mathbf{c}|^2 = 9 + 16 + 25$$

Solve further,

$$|\mathbf{a} + \mathbf{b} + \mathbf{c}|^2 = 50$$

$$|\mathbf{a} + \mathbf{b} + \mathbf{c}| = \sqrt{50}$$

$$= 5\sqrt{2}$$

41. According to the question,

$2x + 3y - 3 = 0$ and $x + ky + 7$ are perpendicular

$$2 \times 1 + 3k = 0$$

$$= \frac{-2}{3}$$

42. The expression for the area of the circle is given by,

$$A = \pi r^2$$

Differentiate on the both sides.

$$\frac{dA}{dr} = 2\pi r$$

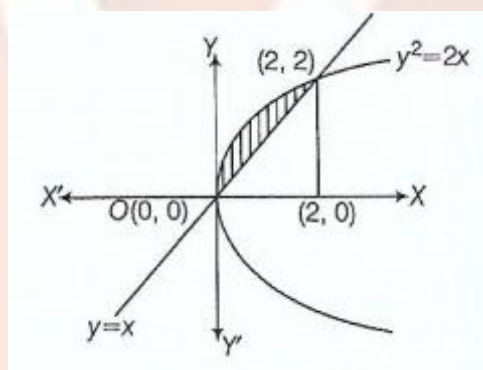
$$\left(\frac{dA}{dr} \right)_{r=2} = 2\pi(2)$$
$$= 4\pi$$

43. Use the following trigonometric identities.

$$\begin{aligned}\tan \theta &= \frac{\pi}{8} \\&= \frac{\sin 2(\pi/8)}{1 + \cos 2(\pi/8)} \\&= \frac{\sin(\pi/4)}{1 + \cos(\pi/4)} \\&= \frac{1}{\sqrt{2} + 1}\end{aligned}$$

44. The given equation of curves are $y^2 = 2x$ and $y = x$.

The solution of the above equation is $x = 2, y = 2$.



The area of the shaded region of the above figure is calculated as,

$$\begin{aligned}\text{Area} &= \int_0^2 (\sqrt{2x} - x) dx \\&= \sqrt{2} \left[\frac{2x^{3/2}}{3} \right]_0^2 - \left[\frac{x^2}{2} \right]_0^2 \\&= \sqrt{2} \left[\frac{2}{3} \times (2)^{3/2} \right] - \left[\frac{2^2}{2} \right] \\&= \frac{8}{3} - 2 \\&= \frac{2}{3} \text{ sq unit}\end{aligned}$$

45. Given value of probability $P(A \cap B) = 7/10$ and $P(B) = 17/20$

The value of $P(A/B)$ is calculated as,

$$\begin{aligned}P(A/B) &= \frac{P(A \cap B)}{P(B)} \\&= \frac{7/10}{17/10} \\&= \frac{14}{17}\end{aligned}$$

46. The coefficient of variation is given by,

$$CV = \frac{\sigma}{x} \times 100$$

$$CV_1 = \frac{\sigma_1}{x_1} \times 100$$

$$60 = \frac{21}{x_1} \times 100$$

$$\overline{x}_1 = 35$$

$$CV_2 = \frac{\sigma_2}{x_2} \times 100$$

$$70 = \frac{16}{x_1} \times 100$$

$$\overline{x}_2 = 22.85$$

47. The probability for the condition “both are Aces” is calculated as,

$$\begin{aligned} \frac{{}^4C_2}{{}^{52}C_2} &= \frac{4 \times 3}{52 \times 51} \\ &= \frac{1}{221} \end{aligned}$$

48. The trigonometric equation is $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$

The solution is as follows,

$$\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$$

$$\sin^{-1} x = \frac{\pi}{2} - \sin^{-1} y$$

$$\sin^{-1} x = \cos^{-1} y$$

$$\sin^{-1} x = \sin^{-1}(\sqrt{1-y^2})$$

Further solve it as,

$$x = \sqrt{1-y^2}$$

$$x^2 = 1 - y^2$$

49. Let assume that
$$I = \int_2^8 \frac{\sqrt{10-x}}{\sqrt{x} + \sqrt{10-x}} dx \dots (1)$$

The following definite integral is solved as,

$$I = \int_2^8 \frac{\sqrt{10-x}}{\sqrt{x} + \sqrt{10-x}} dx$$

Use the following identity $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$

$$I = \int_2^8 \frac{\sqrt{x}}{\sqrt{10-x} + \sqrt{x}} dx \dots (2)$$

Add equation (1) and (2).

$$2I = \int_2^8 \frac{\sqrt{10-x} + \sqrt{x}}{\sqrt{10-x} + \sqrt{x}} dx$$

$$2I = \int_2^8 1 dx$$

$$2I = [x]_2^8$$

$$I = 3$$

50. The converse of the given statement $p \rightarrow q$ is $q \rightarrow p$.

The contrapositive of the given statement is $q \rightarrow p$ is

$$\sim p \rightarrow \sim q.$$

Hence, the contrapositive of the converse of the given statement is “If x is not a prime number, then x is not odd”.

51. The total number of outcomes when two dice are thrown simultaneously is 36

The sample data when the sum of the number is 5

$$\{(1,4), (2,3), (3,2), (4,1)\}.$$

The favourable outcomes is 4

The required probability is calculated as,

$$\begin{aligned}\text{Probability} &= \frac{4}{36} \\ &= \frac{1}{9}\end{aligned}$$

52. The given matrix is $A = \begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}$.

The transpose of the matrix is

$$A^T = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix}$$

$$A + A^T = \begin{bmatrix} \cos 2\theta + \cos 2\theta & -\sin 2\theta + \sin 2\theta \\ \sin 2\theta - \sin 2\theta & \cos 2\theta + \cos 2\theta \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2\cos 2\theta & 0 \\ 0 & 2\cos 2\theta \end{bmatrix}$$

$$2\cos 2\theta = 1$$

Solve further to obtain the value of θ

$$2\cos 2\theta = 1$$

$$2\theta = \frac{\pi}{3}$$

$$\theta = \frac{\pi}{6}$$

53. The given matrix is $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$.

The value of A^2 is calculated as,

$$\begin{aligned} A^2 &= \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix} \end{aligned}$$

The value of $5A$ is calculated as,

$$5A = \begin{bmatrix} 15 & 5 \\ -5 & 10 \end{bmatrix}$$

The value of $A^2 - 5A$ is calculated as,

$$\begin{aligned} A^2 - 5A &= \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix} - \begin{bmatrix} 15 & 5 \\ -5 & 10 \end{bmatrix} \\ &= \begin{bmatrix} -7 & 0 \\ 0 & -7 \end{bmatrix} \\ &= -7I \end{aligned}$$

54. According to the statement of the question,

$$\begin{aligned}
 x\hat{i} + x\hat{j} + x\hat{k} &= 1 \\
 \sqrt{x^2 + x^2 + x^2} &= 1 \\
 3x^2 &= 1 \\
 x &= \pm \frac{1}{\sqrt{3}}
 \end{aligned}$$

55. According to the statement of the question,

$$x = 2 + 3\cos\theta \text{ and } y = 1 - 3\sin\theta$$

$$(x - 2) = 3\cos\theta \dots (1)$$

$$(y - 1) = -3\sin\theta \dots (2)$$

Square and add equation (1) and (2).

$$(x - 2)^2 + (y - 1)^2 = 9$$

Which implies the centre is (2,1) and radius is 3.

56. The expression for the vector equation of plane is given by,

$$\mathbf{r} \cdot \hat{\mathbf{n}} = d$$

The vector equation of the plane is

$$\mathbf{r} \cdot \frac{(2\hat{i} - 3\hat{j} + \hat{k})}{\sqrt{2^2 + 3^2 + 1^2}} = \frac{3}{\sqrt{14}}$$

$$\mathbf{r} \cdot (2\hat{i} - 3\hat{j} + \hat{k}) = 3$$

57. Let assume that the foot of the perpendicular is drawn from origin is α, β, γ to the plane $5y + 8 = 0$.

$$\frac{\alpha - 0}{0} = \frac{\beta - 0}{5} = \frac{\gamma - 0}{0} = \frac{-(0 + 0 + 0 + 8)}{0^2 + 5^2 + 0^2}$$

$$\frac{\alpha}{0} = \frac{\beta}{5} = \frac{\gamma}{0} = \frac{-8}{25}$$

$$\alpha = 0, \beta = \frac{-8}{5}, \gamma = 0$$

Therefore, the coordinates of the foot of perpendicular is

$$\left(0, \frac{-8}{5}, 0\right).$$

58. It is given that $\cos \alpha, \cos \beta, \cos \gamma$ are direction cosines of a vector **a**.

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\frac{1 + \cos 2\alpha}{2} + \frac{1 + \cos 2\beta}{2} + \frac{1 + \cos 2\gamma}{2} = 1$$

$$\cos 2\alpha + \cos 2\beta + \cos 2\gamma = -1$$

59. It is simplified as,

$$\begin{aligned}\sin 1^\circ + \sin 2^\circ + \sin 3^\circ + \dots + \sin 359^\circ &= (\sin 1^\circ + \sin 359^\circ) + \\ &\quad (\sin 2^\circ + \sin 358^\circ) + \dots + \sin 180^\circ \\ &= 0 + 0 + 0 + \dots + 0 \\ &= 0\end{aligned}$$

60. It is given that,

$$x \frac{dy}{dx} - y = x^4 - 3x$$

$$\frac{dy}{dx} - \frac{y}{x} = x^3 - 3$$

Integrating factor (IF) is calculated as,

$$\text{IF} = e^{\int \frac{-1}{x} dx}$$

$$= e^{-\log x}$$

$$= e^{\log \frac{1}{x}}$$

$$= \frac{1}{x}$$